

Virt-Manager E-Core Pinning Guide

Target System: Fedora 44 | Intel Core Ultra 7 270K Plus

To force a Virt-Manager (libvirt) VM to run exclusively on the Efficiency Cores (E-cores) of your Core Ultra 7 270K Plus, you must configure CPU affinity via XML editing. This isolates the VM workload, keeping the host's P-cores free for demanding tasks.

Step 1: Identify your E-core IDs

Recent Linux kernels expose Intel's hybrid topology directly. Intel's internal architectural name for E-cores is "Atom". Open your terminal and run:

```
cat /sys/devices/cpu_atom/cpus
```

Note: Your Ultra 7 270K Plus has 24 total cores (8 P-cores, 16 E-cores). Assuming standard Linux enumeration, your E-cores are likely CPUs 8-23. Use the output from the command above for your exact mapping.

Step 2: Edit the VM XML Configuration

You must edit the VM's XML configuration to pin the virtual CPUs to the physical E-core IDs found in Step 1.

1. Open the VM configuration for editing:

```
virsh edit <your_vm_name>
```

2. Locate the `<vcpu>` tag. Right below it, add a `<cputune>` block to pin both the emulator and each virtual CPU. Below is an example for a 4-vCPU VM using E-cores 8 through 23:

```
<vcpu placement='static'>4</vcpu>
<cputune>
  <!-- Pin the background QEMU emulator threads to the E-core range -->
  <emulatorpin cpuset='8-23' />

  <!-- Pin each virtual CPU to a specific E-core -->
  <vcpupin vcpu='0' cpuset='8' />
  <vcpupin vcpu='1' cpuset='9' />
  <vcpupin vcpu='2' cpuset='10' />
  <vcpupin vcpu='3' cpuset='11' />
</cputune>
```

3. Save and exit your editor.

The next time you start the VM, libvirt will instruct the Linux scheduler to completely isolate the VM's workload to the specified E-cores.